🏆 Infosys SE Elite Crack Vault — 04 Data Structures Arrays Trees and Pointers Elite

🎯 Infosys Systems Engineer (SE) | 📊 39 Questions | 🗓️ 5-6 Nov 2026 Drive | ⚡ 100% Pure SE Verified

Problem 01 • Data Structures, Trees & Pointers

🔥 MUST-SOLVE

ID: SEC03_EBOOK_Q004 | Source: 1. Pseudo Codes Technical ebook.pdf | Level: Infosys SE Standard

/ Code Snippet

```
Where are placed the list of
processes that are prepared to be
executed and waiting?
```

- a) Job queue
- b) Ready queue
- c) Execution queue
- d) Process queue

✓ **Verified Answer:** Option B

Problem 02 • Recursion & Function Call Stacks

🔥 MUST-SOLVE

ID: SEC03_EBOOK_Q010 | Source: 1. Pseudo Codes Technical ebook.pdf | Level: Infosys SE Standard

/ Code Snippet

```
What does the expression float a =
35 / 0 return?
```

- a) 0
- b) Not a Number
- c) Infinity
- d) Run time exception

✓ **Verified Answer:** Option C

Problem 03 • Data Structures, Trees & Pointers

MUST-SOLVE

ID: SEC03_EBOOK_Q012 | Source: 1. Pseudo Codes Technical ebook.pdf | Level: Infosys SE Standard

/ Code Snippet

```
Construct post order Transversal
for the constructed tree with the
given
In-Order
and
Preorder
Traversals:
In-Order: 9 7 10 6 8
Pre-Order: 6 7 9 10 8
```

- a) 9 10 7 8 6
- b) 9 7 6 10 8
- c) 8 9 10 7 6
- d) 9 8 10 7 6

✓ **Verified Answer:** Option A

Problem 04 • Recursion & Function Call Stacks

MUST-SOLVE

ID: SEC03_EBOOK_Q015 | Source: 1. Pseudo Codes Technical ebook.pdf | Level: Infosys SE Standard

/ Code Snippet

```
What will be the output of this
program?
#include<stdio.h>
int main() {
    int i = 5;
    printf(“%d”, i = ++i == 6);
return 0;
}
```

- a) 2
- b) 6
- c) 4
- d) 1

✓ **Verified Answer:** Option D

Problem 05 • Recursion & Function Call Stacks

MUST-SOLVE

ID: SEC03_EBOOK_Q017 | Source: 1. Pseudo Codes Technical ebook.pdf | Level: Infosys SE Standard

/ Code Snippet

```
What will be the output of this
program?
#include<stdio.h>
enum flg{a, b, c};
enum glf{c, e, f};
main()
{
  enum flg h;
    h = b;
    printf(“%d”, h);
   return 0;
}
```

- a) 1
- b) error: redeclaration of an
- c) h
- d) 3

✓ **Verified Answer:** Option B

Problem 06 • Data Structures, Trees & Pointers

MUST-SOLVE

ID: SEC03_EBOOK_Q021 | Source: 1. Pseudo Codes Technical ebook.pdf | Level: Infosys SE Standard

/ Code Snippet

```
If the sequence of operations are
performed on a stack, the sequence
of pop will be?
Push(1), Push(2), Pop(), Push(1),
Push(2), Pop(), Pop(), Pop(),
Push(2), Pop().
```

- a) 2 2 1 1 2
- b) 2 2 1 2 2
- c) 2 1 2 2 1
- d) 2 1 2 2 2

✓ **Verified Answer:** Option A

Problem 07 • Data Structures, Trees & Pointers

MUST-SOLVE

ID: SEC03_EBOOK_Q023 | Source: 1. Pseudo Codes Technical ebook.pdf | Level: Infosys SE Standard

/ Code Snippet

```
The concatenation of two lists is to
be performed in O(1) time. Which of
the following implementation of a
list should be used?
```

- a) Singly linked list
- b) Doubly linked list
- c) Circular doubly linked list
- d) Array implementation of list

✓ **Verified Answer:** Option C

Problem 08 • Data Structures, Trees & Pointers

MUST-SOLVE

ID: SEC03_EBOOK_Q024 | Source: 1. Pseudo Codes Technical ebook.pdf | Level: Infosys SE Standard

/ Code Snippet

```
Which of the following sequence
can be obtained in the output (in
same
order)
using
a
stack
assuming that the input is in the
sequence 1,2,3,4,5 order?
```

- a) 3,4,5,1,2
- b) 3,4,5,2,1
- c) 1,5,2,3,4
- d) 5,4,3,1,2

✓ **Verified Answer:** Option B

Problem 09 • Recursion & Function Call Stacks

MUST-SOLVE

ID: SEC03_EBOOK_Q026 | Source: 1. Pseudo Codes Technical ebook.pdf | Level: Infosys SE Standard

/ Code Snippet

```
The value of j at the end of the
execution of the following pseudo
code is ?
Integer incr(Integer i)
Static Integer count = 0
count = count +i
return count
Main()
Integer i,j
for(start i from 0 to 4)
j= incr(i)
end for
```

- a) 10
- b) 4
- c) 6
- d) 7

✓ **Verified Answer:** Option A

Problem 10 • Recursion & Function Call Stacks

MUST-SOLVE

ID: SEC03_EBOOK_Q027 | Source: 1. Pseudo Codes Technical ebook.pdf | Level: Infosys SE Standard

/ Code Snippet

```
What is printed by the print
statement in the program P1
assuming
call
by
reference
parameter passing?
program P1()
{
X = 10
Y = 3
Func1(Y,X,X)
Print X
Print Y
}
Func1(X,Y,Z)
{
Y= Y+4
Z= X+Y+Z
}
```

- a) 10,3
- b) 31,3
- c) 27,7
- d) None of these

✓ **Verified Answer:** Option B

Problem 11 • Recursion & Function Call Stacks

MUST-SOLVE

ID: SEC03_EBOOK_Q028 | Source: 1. Pseudo Codes Technical ebook.pdf | Level: Infosys SE Standard

/ Code Snippet

```
The value returned by the following
function for f(1) is?
int f(int n)
{
static int i =1;
if (n>=5) return n;
n=n + i ;
i++;
return f(n);
}
```

- a) 5
- b) 6
- c) 7
- d) 8

✓ **Verified Answer:** Option C

Problem 12 • Data Structures, Trees & Pointers

MUST-SOLVE

ID: SEC03_EBOOK_Q034 | Source: 1. Pseudo Codes Technical ebook.pdf | Level: Infosys SE Standard

/ Code Snippet

```
Post order traversal of a tree is: V R
T S U Q P Which of the following
vertices will occupy positions 2 and
3?
```

- a) S, R
- b) F, G
- c) U, T
- d) X, Y

✓ **Verified Answer:** Option A

Problem 13 • Data Structures, Trees & Pointers

MUST-SOLVE

ID: SEC03_EBOOK_Q035 | Source: 1. Pseudo Codes Technical ebook.pdf | Level: Infosys SE Standard

/ Code Snippet

```
Which cannot be the number of
vertices in a full binary tree?
```

- a) 3
- b) 7
- c) 15
- d) 30

✓ **Verified Answer:** Option D

Problem 14 • Data Structures, Trees & Pointers

MUST-SOLVE

ID: SEC03_EBOOK_Q036 | Source: 1. Pseudo Codes Technical ebook.pdf | Level: Infosys SE Standard

/ Code Snippet

```
Sam sorts an array by repeatedly
swapping the adjacent elements if
they are in wrong order. Identify the
sorting technique used.
```

- a) Merge Sort
- b) Selection Sort
- c) Bubble Sort
- d) Insertion Sort

✓ **Verified Answer:** Option C

Problem 15 • Data Structures, Trees & Pointers

CORE SE

ID: SEC03_EBOOK_Q039 | Source: 1. Pseudo Codes Technical ebook.pdf | Level: Infosys SE Standard

/ Code Snippet

```
Consider a stack of integers. The
stack is allocated 7 memory cells:
STACK: 12, 34, 66, 79, _, _, _.
Now
consider
the
following
operations:
A. PUSH (STACK, 8)
B. PUSH (STAC, 54)
C. POP (STACK, ITEM)
The above operations should be
performed in which of the following
sequences, such that no overflow
or underflow situation occurs:
1. B-> A-> C
2. A-> B-> C
3. C-> B-> A
```

- a) Only 1
- b) Any of them
- c) Only 2
- d) Only 3

✓ **Verified Answer:** Option B

Problem 16 • Recursion & Function Call Stacks

CORE SE

ID: SEC03_EBOOK_Q040 | Source: 1. Pseudo Codes Technical ebook.pdf | Level: Infosys SE Standard

/ Code Snippet

```
Find the output of the following
pseudo code, if function is called for
x=1 and y=4.
Integer fun (Integer x, Integer y)

If (y > x)


return 1+ fun (x, y-1)

End if

return x + 1
End function fun()
```

- a) 5
- b) -15
- c) 14
- d) 22

✓ **Verified Answer:** Option A

Problem 17 • Recursion & Function Call Stacks

CORE SE

ID: SEC03_EBOOK_Q046 | Source: 1. Pseudo Codes Technical ebook.pdf | Level: Infosys SE Standard

/ Code Snippet

```
#include<stdio.h>
int main()
{
int a = 11, b=4, c=3;
b!= !a;
c = !!a;
printf("%d %d", b,c);
return 0;
}
```

- a) 1,1
- b) 2,1
- c) 3,1
- d) 4,1

✓ **Verified Answer:** Option D

Problem 18 • Data Structures, Trees & Pointers

CORE SE

ID: SEC03_EBOOK_Q047 | Source: 1. Pseudo Codes Technical ebook.pdf | Level: Infosys SE Standard

/ Code Snippet

```
For a B- tree of order 6, to be a
multi-way search tree, each non
root node has at least how many
Non empty Children?
```

- a) 3
- b) 4
- c) 5
- d) 6

✓ **Verified Answer:** Option A

Problem 19 • Data Structures, Trees & Pointers

CORE SE

ID: SEC03_EBOOK_Q050 | Source: 1. Pseudo Codes Technical ebook.pdf | Level: Infosys SE Standard

/ Code Snippet

```
Find out the array representation of
the given min-heap, if the value 25
is inserted in it.
10,15,30,40,50,100
```

- a) 10,15,25,50,40,100,30
- b) 10,15, 25,40,50,100,30
- c) 15,10,25,40,50,100,30
- d) 10,15,25,40,50,30,100

✓ **Verified Answer:** Option B

Problem 20 • Data Structures, Trees & Pointers

CORE SE

ID: SEC03_EBOOK_Q051 | Source: 1. Pseudo Codes Technical ebook.pdf | Level: Infosys SE Standard

/ Code Snippet

```
Which of the following is not an
array representation of max heap?
1. 31, 27, 23, 17, 13, 11
2. 10, 5, 3, 2, 4
3. 12, 15, 17, 18, 19
```

- a) Only 1
- b) Only 3
- c) Only 2 and 3
- d) Only 1 and 3

✓ **Verified Answer:** Option B

Problem 21 • Recursion & Function Call Stacks

CORE SE

ID: SEC03_EBOOK_Q053 | Source: 1. Pseudo Codes Technical ebook.pdf | Level: Infosys SE Standard

/ Code Snippet

```
What will be the output of following
pseudocode?
function val: integer
return (0)
end-function
function main()
int x=val(10)
display x
end-function
```

- a) 0
- b) 1
- c) 2
- d) 3

✓ **Verified Answer:** Option A

Problem 22 • Data Structures, Trees & Pointers

CORE SE

ID: SEC03_EBOOK_Q056 | Source: 1. Pseudo Codes Technical ebook.pdf | Level: Infosys SE Standard

/ Code Snippet

```
What will be the output of following
pseudocode?
Set integer array=[2,4,0,1,6,0]
Set integer res=[2]/array[4];
display res
```

- a) 0
- b) 1
- c) 2
- d) 3

✓ **Verified Answer:** Option A

Problem 23 • Recursion & Function Call Stacks

CORE SE

ID: SEC03_EBOOK_Q068 | Source: 1. Pseudo Codes Technical ebook.pdf | Level: Infosys SE Standard

/ Code Snippet

```
Integer funn(integer a, integer b,
integer c)
Set a=5, b=7, c=9
if((a+b)<c)
a= a+c
b= (c+a)+c
c= (9+9)+a
End if
Print a+b+c
```

- a. 20
- b. 21
- c. 22
- d. 23

✓ **Verified Answer:** Option B

Problem 24 • Data Structures, Trees & Pointers

CORE SE

ID: SEC03_EBOOK_Q097 | Source: 1. Pseudo Codes Technical ebook.pdf | Level: Infosys SE Standard

/ Code Snippet

```
Which of the following data
structure is used to convert
expression from one form to
another?
```

- a. Graph
- b. Stack
- c. Linked list
- d. Queue

✓ **Verified Answer:** Option B

Problem 25 • Data Structures, Trees & Pointers

CORE SE

ID: SEC03_EBOOK_Q100 | Source: 1. Pseudo Codes Technical ebook.pdf | Level: Infosys SE Standard

/ Code Snippet

```
Find the pre order of the
given tree.
  A

        /         \
     B
          C
   /   \            /   \
D     E        F    G
          \        \
H       I
```

- a. ABDEHCFIG
- b. BASUEKDHS
- c. AABBLIKDMSD
- d. CAMPUSMONK

✓ **Verified Answer:** Option A

Problem 26 • Recursion & Function Call Stacks

CORE SE

ID: SEC03_EBOOK_Q111 | Source: 1. Pseudo Codes Technical ebook.pdf | Level: Infosys SE Standard

/ Code Snippet

```
Find the output of the code
snippet.
#include<stdio.h>
int main()
{
int i=0;
for (; i<=5; i++);
printf("%d", i);
return 0;
}
```

- a. 5
- b. 0 1 2 3 4 5
- c. 6
- d. 0 1 2 3 4

✓ **Verified Answer:** Option C

Problem 27 • Recursion & Function Call Stacks

CORE SE

ID: SEC03_EBOOK_Q112 | Source: 1. Pseudo Codes Technical ebook.pdf | Level: Infosys SE Standard

/ Code Snippet

```
Find the output of the code
snippet.
#include<stdio.h>
int main()
{
unsigned int i=65535;
while(i++ != 0)
printf("%d", ++i);
return 0;
}
```

- a. No output printed
- b. 012 3 65535
- c. 012 3.....32767 -32766 -
- d. Inifinite loop

✓ **Verified Answer:** Option D

Problem 28 • Recursion & Function Call Stacks

CORE SE

ID: SEC03_EBOOK_Q113 | Source: 1. Pseudo Codes Technical ebook.pdf | Level: Infosys SE Standard

/ Code Snippet

```
Find the output of the code
snippet.
int main()
{
int a=500,b=100, c;
if(!a>=400)
b=300;
c=200;
printf("%d%d",b,c);
return 0;
}
```

- a. 300 200
- b. 100 200
- c. 100 Garbage value
- d. 300 Garbage value

✓ **Verified Answer:** Option B

Problem 29 • Data Structures, Trees & Pointers

CORE SE

ID: SEC03_EBOOK_Q116 | Source: 1. Pseudo Codes Technical ebook.pdf | Level: Infosys SE Standard

/ Code Snippet

```
If the sequence of
operations are performed on a
stack, the sequence of pop will be?
Push(1), Push(2), Pop(), Push(1),
Push(2), Pop(), Pop(), Push(2),
Pop().
```

- a. 2 2 1 1 2
- b. 2 2 1 2 2
- c. 2 1 2 2 1
- d. 2 1 2 2 1

✓ **Verified Answer:** Option B

Problem 30 • Data Structures, Trees & Pointers

CORE SE

ID: SEC03_EBOOK_Q117 | Source: 1. Pseudo Codes Technical ebook.pdf | Level: Infosys SE Standard

/ Code Snippet

```
Consider the following
operations performed on a stack of
size 5. Push(a), Pop(), Push(b),
Push(c), Pop(), Push(d), Pop(),
Pop(), Push(e) Which of the
following statements is correct?
```

- a. Underflow occurs
- b. Stack operations are performed
- c. Overflow occurs
- d. None

✓ **Verified Answer:** Option B

Problem 31 • Data Structures, Trees & Pointers

CORE SE

ID: SEC03_EBOOK_Q118 | Source: 1. Pseudo Codes Technical ebook.pdf | Level: Infosys SE Standard

/ Code Snippet

```
Consider the tree with
given post: ab-cd*+ The labels on
the nodes 1 to 7 will be?2
```

- a. a, -, b, *, c, +, d
- b. +, -, *, a, b, c, d
- c. a, b, c, d, -, *, +
- d. -, a, b, +, *, c, d

✓ **Verified Answer:** Option B

Problem 32 • Data Structures, Trees & Pointers

CORE SE

ID: SEC03_EBOOK_Q120 | Source: 1. Pseudo Codes Technical ebook.pdf | Level: Infosys SE Standard

/ Code Snippet

```
Which of the following
relation is correct for a binary
search tree?
```

- a. Parent node> Right node>
- b. Right node> Parent node>
- c. Right node> Left node>
- d. Parent node> Left

✓ **Verified Answer:** Option B

Problem 33 • Recursion & Function Call Stacks

CORE SE

ID: SEC03_EBOOK_Q125 | Source: 1. Pseudo Codes Technical ebook.pdf | Level: Infosys SE Standard

/ Code Snippet

```
Find the output of the
following code.
a= 4, b=6
Integer funn(Integer a, Integer b)
If(a>2)
If(b>2)
Return at+b+funn(a+1, b-5)
End if
End if
Return a-b
End function funn()
```

- a. 14
- b. 10
- c. 2
- d. 5

✓ **Verified Answer:** Option A

Problem 34 • Recursion & Function Call Stacks

CORE SE

ID: SEC03_EBOOK_Q129 | Source: 1. Pseudo Codes Technical ebook.pdf | Level: Infosys SE Standard

/ Code Snippet

```
Find the output of the
following code.
int main()
{
int y = 128;
const int x=y;
y=120;
printf(“%d\n”, x);
return 0;
}
```

- a. 0
- b. 128
- c. Garbage Value
- d. Error

✓ **Verified Answer:** Option B

Problem 35 • Recursion & Function Call Stacks

CORE SE

ID: SEC03_EBOOK_Q131 | Source: 1. Pseudo Codes Technical ebook.pdf | Level: Infosys SE Standard

/ Code Snippet

```
Find the output of the
following code.
#includex<stdio.h>
int main()
{
Printf(“\\n”);
return 0;
}
```

- a. \n
- b. \n
- c. Error
- d. n

✓ **Verified Answer:** Option B

Problem 36 • Recursion & Function Call Stacks

CORE SE

ID: SEC03_EBOOK_Q132 | Source: 1. Pseudo Codes Technical ebook.pdf | Level: Infosys SE Standard

/ Code Snippet

```
Find the complete of the
following code where the output is
8,14.
#include<stdio.h>
int main()
{
int p=6, q=6;
intr=6*(_____________);
printf (“%d%d”, p,r);
return 0;
}
```

- a. (p + q + p++)
- b. (p++ + q + p)
- c. (p++ +q + p++)
- d. none

✓ **Verified Answer:** Option C

Problem 37 • Recursion & Function Call Stacks

CORE SE

ID: SEC03_EBOOK_Q133 | Source: 1. Pseudo Codes Technical ebook.pdf | Level: Infosys SE Standard

/ Code Snippet

```
Find the output of the
following code.
#include<stdio.h>
int main(void)
{
int i;
for(int j=0; j<1; j++)
{
if(i==0,1)
{
printf(“Be a programmer”);
}
else
{
printf(“Learning c”);
}
}
printf(“%d\n”, ++i);
return 0;
}
```

- a. Be a programmer1
- b. Be a programmer0
- c. Learning c1
- d. Learning c0

✓ **Verified Answer:** Option A

Problem 38 • Recursion & Function Call Stacks

CORE SE

ID: SEC03_EBOOK_Q135 | Source: 1. Pseudo Codes Technical ebook.pdf | Level: Infosys SE Standard

/ Code Snippet

```
find the output of the
following code.
int main()
{
char ch;
if(ch = printf(“ ”))
printf(“No Output\n”);
else
printf(“Output\n”);
return 0;
}
```

- a. No Output
- b. Output
- c. No Output Output
- d. Error

✓ **Verified Answer:** Option B

Problem 39 • Recursion & Function Call Stacks

CORE SE

ID: SEC03_EBOOK_Q136 | Source: 1. Pseudo Codes Technical ebook.pdf | Level: Infosys SE Standard

/ Code Snippet

```
find the output of the
following code.
#include<stdio.h>
int main(void)
{
int a=10, x;
X= a-- + +445
printf (“%d”,x);
return 0;
}
```

- a. 19
- b. 20
- c. 22
- d. 23

✓ **Verified Answer:** Option B